# Understanding and Reducing Missed GP Appointments: A Data-Driven Analysis of NHS Capacity and Resource Utilisation- Technical Report

**By: Reshma Pandit**

**Background:** The National Health Service (NHS) would like to understand and reduce the financial and operational impact of missed GP appointments across England. Recent reports suggest that some regions experience higher rates of missed appointments, uneven staff capacity, and under-utilised resources. These inefficiencies not only strain NHS budgets but also delay care for patients who need timely access.

The NHS needs a data-informed approach to decide how to reduce missed appointments and optimise current operations. This analysis hence focusses on understanding staff capacity, appointment utilisation, and potential drivers of missed appointments. The findings will help the NHS identify where improvements in planning, communication, or capacity management could deliver the greatest impact, both financially and socially.

The analysis focuses on KPIs that reflect appointment status and utilisation trends across regions, contexts, and service types.

**Business questions:**

1. Has there been adequate staff and capacity in the networks?
2. What was the actual utilisation of resources?
3. How could we reduce missed appointments to optimise current operations?

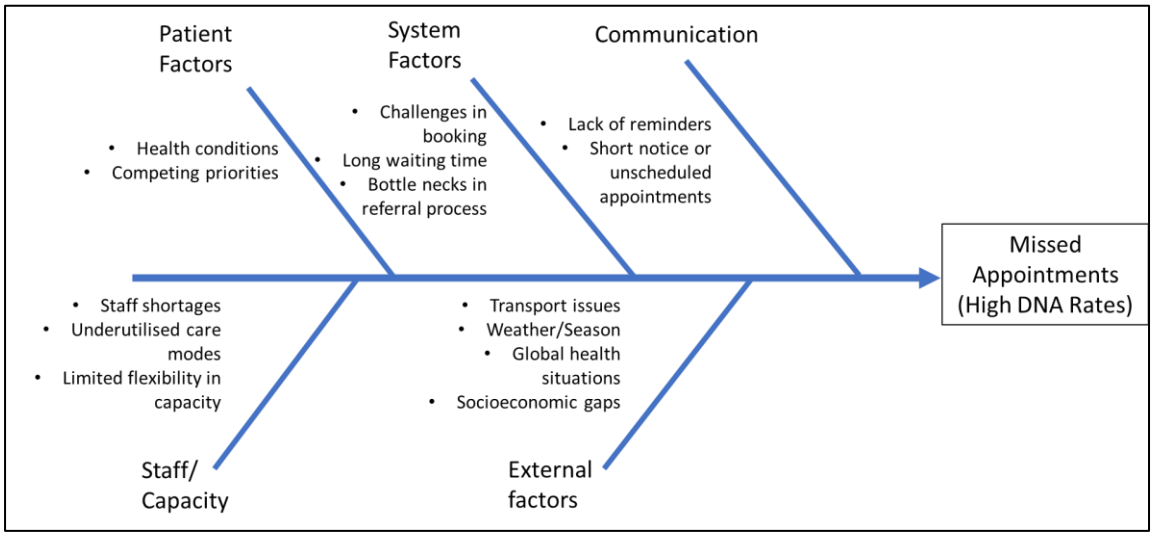
**Analytical questions:**

The data is approached with the following questions to inform operational decisions on NHS capacity and allocation of resources:

1. Quantify appointment activity across regional and service setting to establish the baseline NHS utilisation in the networks.
2. Measure variation in appointment volumes over time to detect seasonal patterns that could indicate under or over-utilisation of NHS capacity.
3. Assess utilisation across healthcare professional types and service settings to identify care types that experience consistent pressure.
4. Calculate and compare Did Not Attend (DNA) rates across waiting periods and appointment modes to highlight factors that contribute to missed appointments, and to support targeted interventions and communication strategies.
5. Analyse public sentiment from Twitter data to uncover recurring themes about the NHS system that may influence attendance behaviour.

**Analytical approach:**

The exploratory analysis of this dataset has been approached through the **5-Whys framework**. The fishbone diagram of various factors is as given below:




**Data Source:**

NHS has provided data in the following files: actual_duration.csv (=ad), appointments_regional.csv (=ar), national_categories.xlsx (=nc) and tweets.csv (=tw).

**Libraries**:

The libraries used for this analysis are as below:

```
# Install missing packages
!pip install textblob wordcloud

# Import libraries
import pandas as pd
import numpy as np
from datetime import datetime
import matplotlib.pyplot as plt
import seaborn as sns
from matplotlib.colors import LinearSegmentedColormap

# Text & sentiment analysis
import re
import string
from textblob import TextBlob
from nltk.sentiment.vader import SentimentIntensityAnalyzer
from wordcloud import WordCloud
from collections import Counter
import nltk
nltk.download('vader_lexicon')

# Utility
import warnings
warnings.filterwarnings('ignore')
```

**Colour palette:**

For consistent styling of plots, the NHS colour palette was obtained and set up.

**Functions**:

Considering repeated usage of certain actions, the following functions were defined:

1. validate_data(df)
2. clean_text(s)
3. clean_label(text)
4. generate_standard_bar_chart
5. generate_standard_line_plot
6. clean_tweet(text)
7. extract_hashtaga(text)

**Ingestion & schema validation:**

The .csv and .xlsx datasets were imported into Pandas DataFrames for analysis. Initial validation steps included verifying record counts, expected column names, and data types to ensure schema integrity. Data fields were coerced to appropriate formats; for example, converting date fields to datetime objects to prevent computational errors and enable reliable time-series analysis.

**Standardisation & cleaning:**

Applied a clean_text() function to strip whitespace and convert categorical labels to a consistent case, improving groupby consistency. Used validate_data function to determine unique values per column, summary statistics, no. of duplicate rows, missing values and potential outliers (using IQR method). Unique categories in key fields were checked to ensure no spelling errors or repeat values due to formatting/data errors. Unknown values were investigated. Record counts and date ranges of the datasets were determined to ensure data is within expected range.

Dataset specific points:

For ad:

Outliers in count_of_appointments were retained as they most likely reflect high-volume locations rather than errors. Around 4% of records had “Unknown” appointment statuses across all 106 ICBs, suggesting incomplete data capture.

For ar:

Duplicate records were retained because categorical aggregation lacked unique identifiers. Outliers and “Unknown” statuses were treated as valid observations as in ad.

For nc:

Unmapped and extreme values were retained, consistent with the above approach.

For tw:

Null hashtags were replaced with empty strings, as tweets commonly omit hashtags.

**Feature engineering:**

Derived year_month for monthly trends. Computed DNA_rate as Pivot["dna_rate"] = (pivot["dna"] / pivot["total_appointments"]) * 100 to assess proportion of DNA appointments from all appointments. Created dictionary mapping region_ons_code to readable names for visualising data across regions.

**Exploratory aggregation & visuals:**

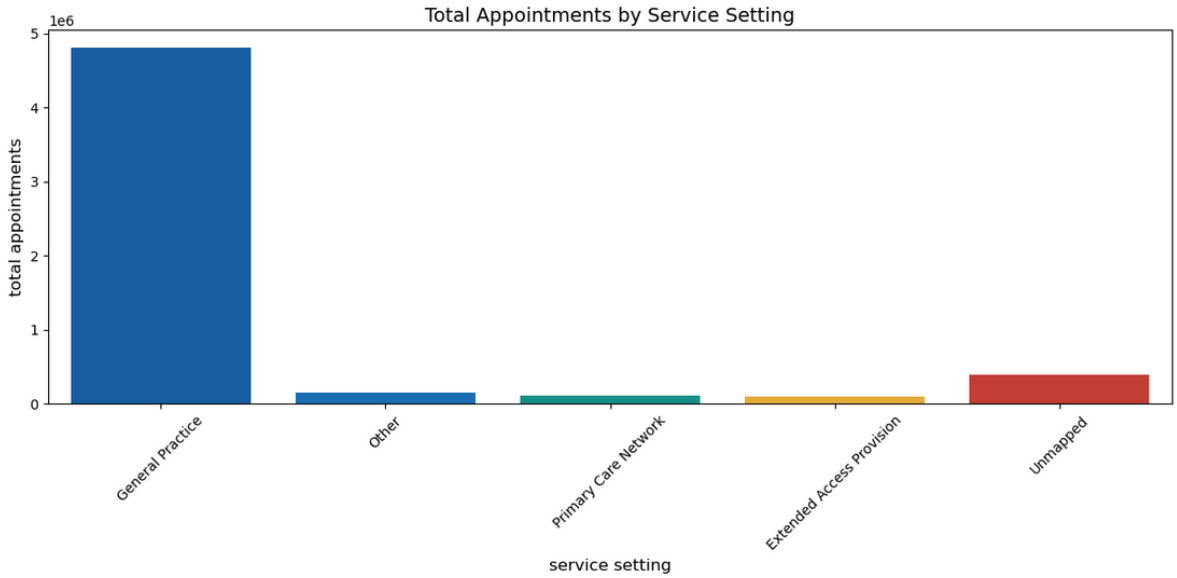
Exploratory aggregation and visualisation examined trends in appointment volumes, attendance behaviour, and DNA rates across regions and service settings. Boxplots identified outliers; bar charts compared locations; line plots showed temporal trends; and heatmaps explored multivariate relationships. Word clouds and sentiment scores from Twitter data provided qualitative context to patient attitudes. Used groupby/pivot_table for heatmaps for visualisations, such as for appointment_status.

**Trends and patterns from Exploratory analysis:**

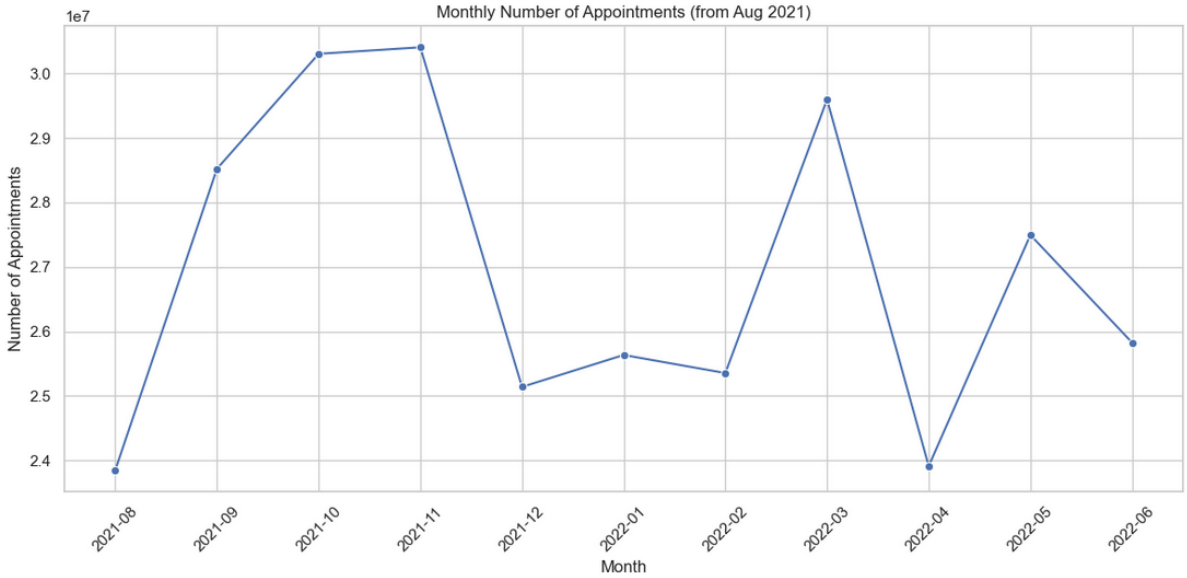
1. **Capacity and Network Utilisation:**

   General Practice (GP) is the most utilised service setting as seen below.

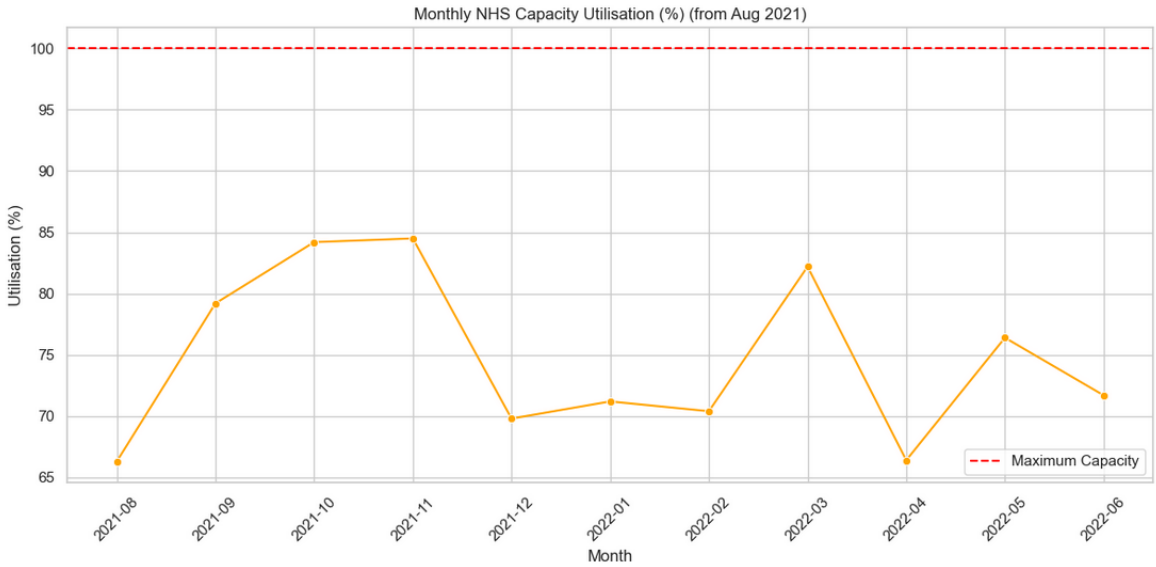
| service_setting | number_of_records | total_appointments |
|---|---|---|
| General Practice | 2104 | 4804239 |
| Other | 1318 | 152897 |
| Primary Care Network | 1272 | 109840 |
| Extended Access Provision | 1090 | 98159 |
| Unmapped | 152 | 391106 |




Appointment volumes ranged from 23-31 million per month, peaking in October 2021 and March 2022, coinciding with schools reopening after lifting of COVID-19 restrictions in 2021 and after lifting of all COVID-19 restrictions in February 2022.

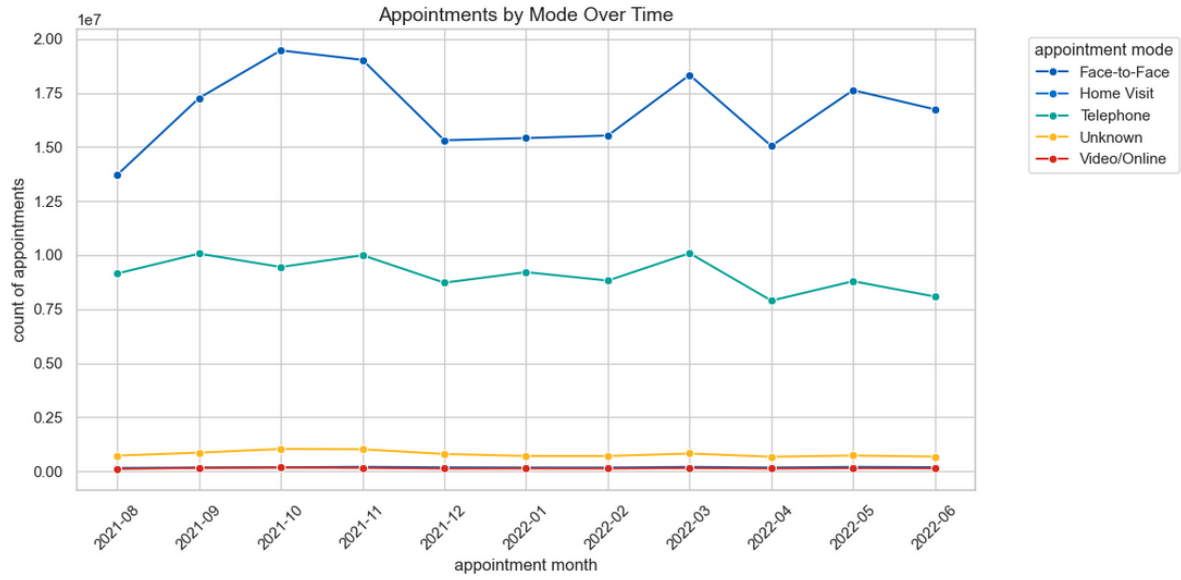



Despite high demand, utilisation reached up to ~85% of planned capacity, implying that current system capacity is largely adequate but could be better balanced across time and regions.




**2. Resource Utilisation and Efficiency**

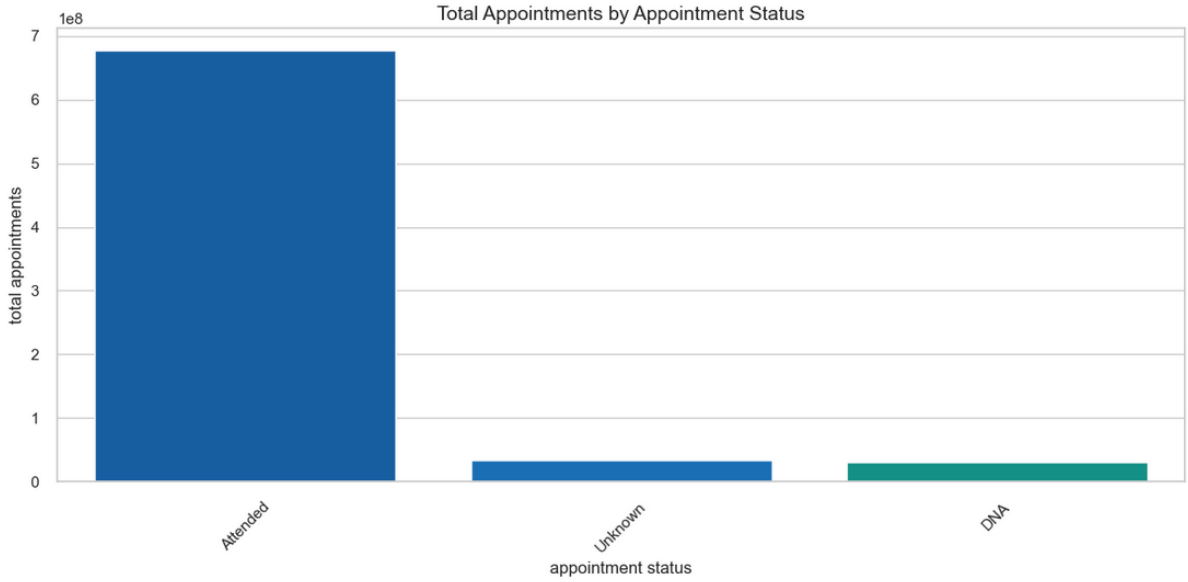
Face-to-face appointments remain the dominant mode, absorbing most of the variability in demand (October–November 2021 and March 2022). Telephonic and video consultations remain steady and under-utilised, representing an opportunity to redistribute workload and improve flexibility in service delivery, especially in peak demand scenarios.



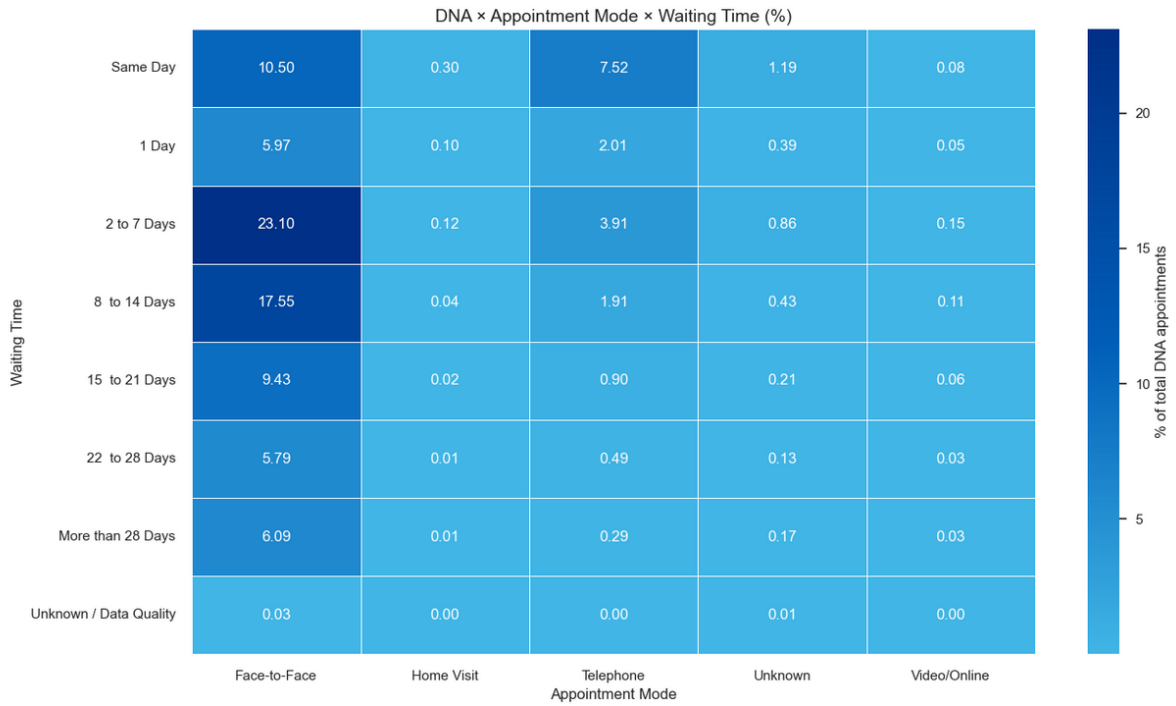

### 3. Reducing Missed Appointments (DNA Rates)

The overall DNA rate averages ~4%, equivalent to over 30 million missed appointments nationally for this date range of Jan 2020 to June 2022. A large share of records have "unknown" appointment status (~4%), limiting full operational visibility. Improving data quality here would enhance planning and measurement.

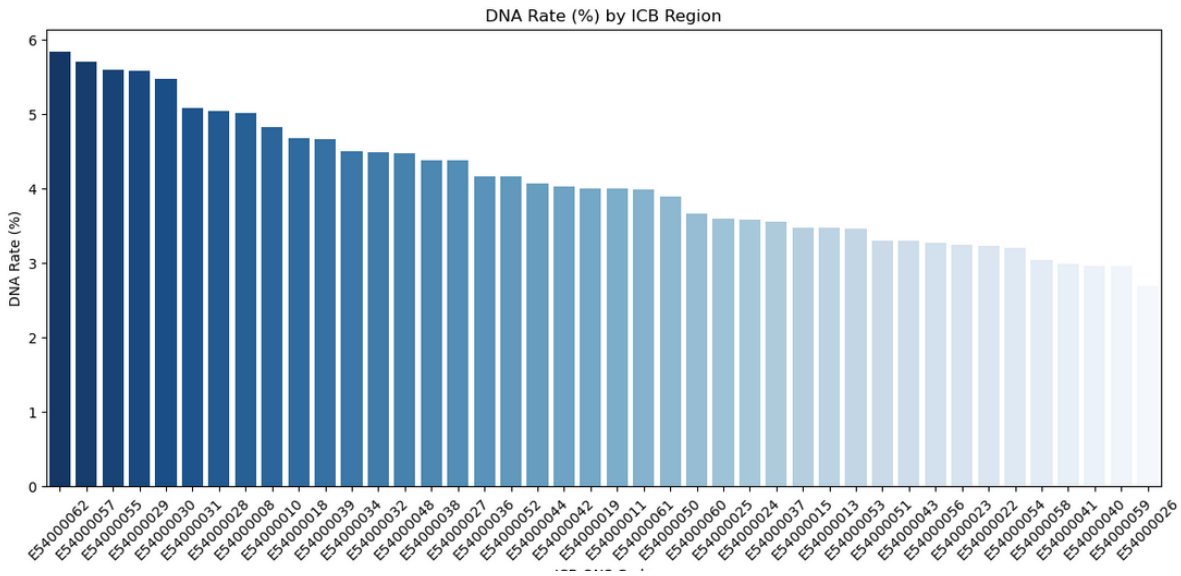
| | appointment_status | total_appointments | record_count | percent_of_total |
|---|---|---|---|---|
| 0 | Attended | 677755876 | 232137 | 91.24 |
| 2 | Unknown | 34137416 | 201324 | 4.60 |
| 1 | DNA | 30911233 | 163360 | 4.16 |




Face-to-face and same-day appointments show higher DNA rates, possibly due to travel barriers or short notice. Video/online and home visits show the lowest DNA rates, suggesting that flexible appointment options and advance scheduling improve attendance.

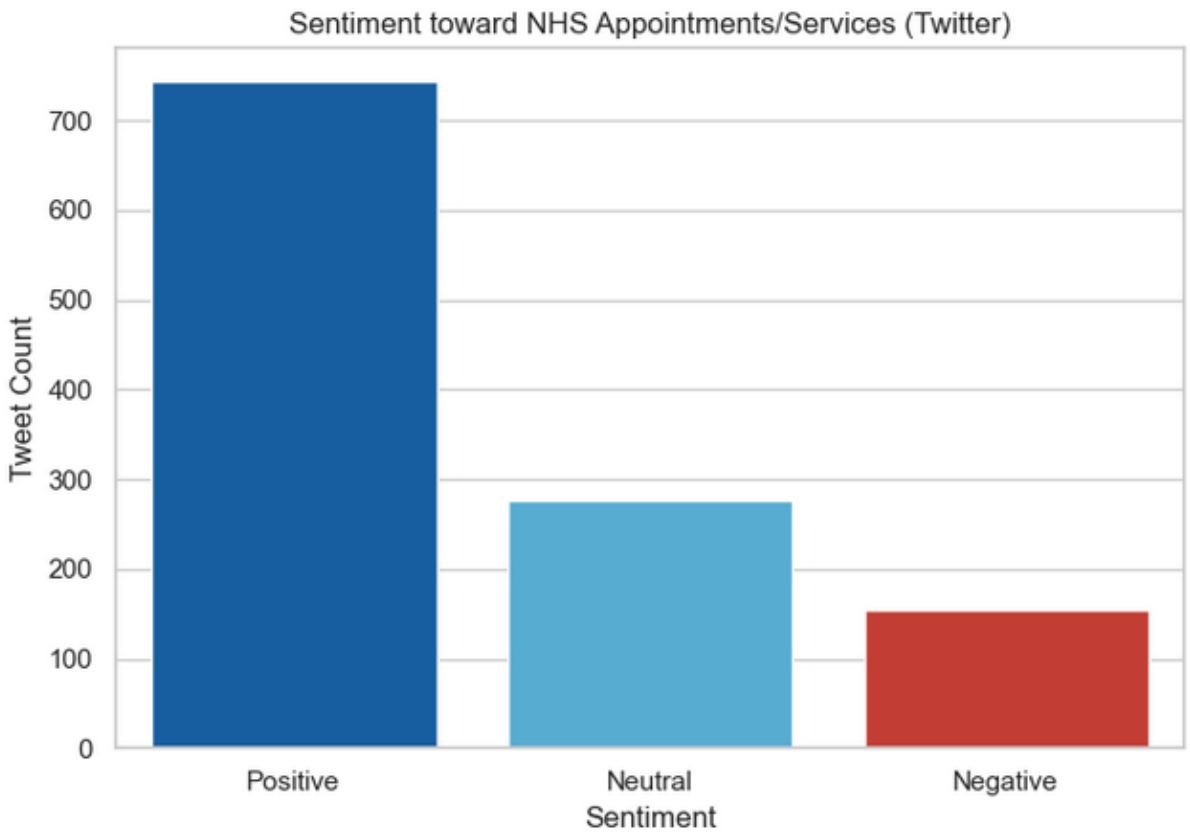


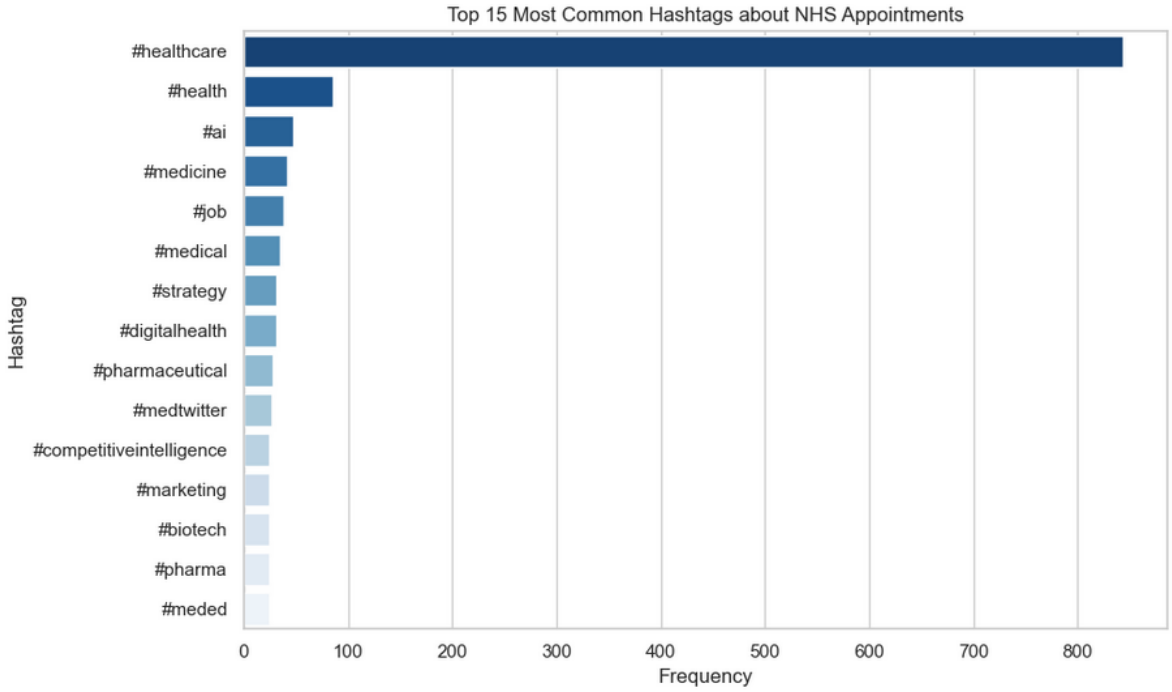

DNA rates vary more than twofold across ICBs (from ~2.7% to ~5.8%). Highest DNA: Black Country (E54000062), Greater Manchester (E54000057), Birmingham & Solihull (E54000055). Lowest DNA: Mid & South Essex (E54000026), Northamptonshire (E54000059), Bath & North East Somerset (E54000040). These differences highlight regional disparities and target opportunities for behavioural or operational interventions along with learning opportunities to reduce DNA appointments.




### 4. Public Sentiment and Communication

Twitter sentiment about the NHS was broadly positive, with minimal mentions of missed appointments or waiting-time complaints; implying that public perception is not a major barrier to attendance.


Sentiment toward NHS Appointments/Services (Twitter)
Tweet Count
0
100
200
300
400
500
600
700
Positive
Neutral
Negative
Sentiment


Top 15 Most Common Hashtags about NHS Appointments
Hashtag
#healthcare
#health
#ai
#medicine
#job
#medical
#strategy
#digitalhealth
#pharmaceutical
#medtwitter
#competitiveintelligence
#marketing
#biotech
#pharma
#meded
0
100
200
300
400
500
600
700
800
Frequency

**Key Technical limitations:**

1. Many appointments are “Unmapped” or “Unknown” which if available could have confirmed the various observed trends.
2. Data on capacity/staffing is lacking, resulting in a limited interpretation of staff capacity for the workload.

**Recommendations and Further actions:**

1. Strengthen Data Infrastructure: Standardise appointment data entry and automate validation.
2. Continuous Performance Monitoring: Track utilisation and DNA rates by ICB, mode, and month. Introduce performance dashboards for real-time utilisation tracking.